

CIVL4340

Wind Engineering

Week 11 & 12: Dynamic wind effects

Overview

Dynamic response to wind

- Random vibration
- Forced (sinusoidal) vibration
- Wind-induced vibration

Reading

- Holmes, Chapters 5 (9).
- Simiu and Yao, Chapter 11

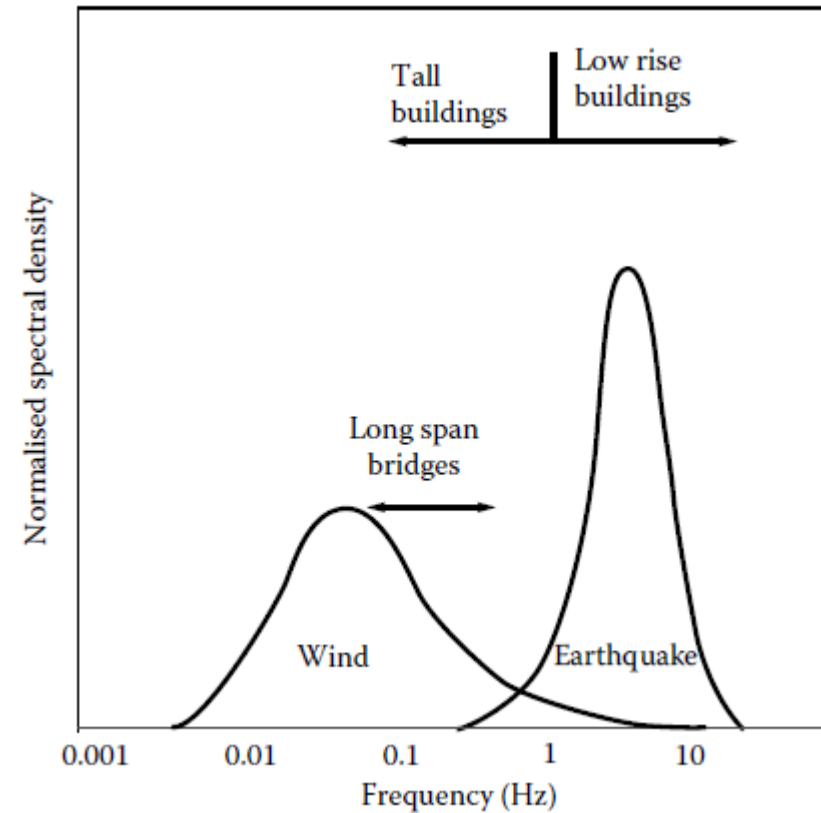
Dynamic response

When structures (or components of a structure) have **natural frequencies** less than around 1hz they may have issues with dynamic response (vibration) due to wind forces.

Structures resist vibration through

- Mass (inertial forces)
- Damping or energy absorption
- Stiffness

Wind vs earthquake

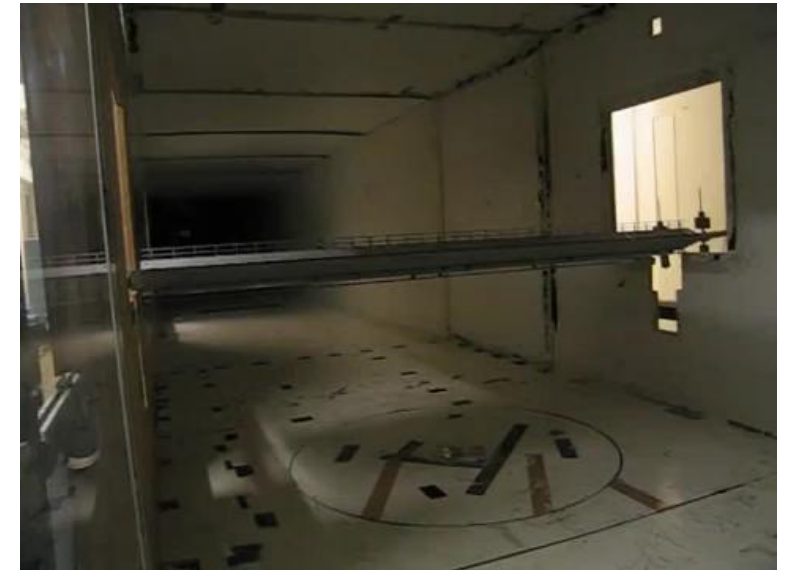
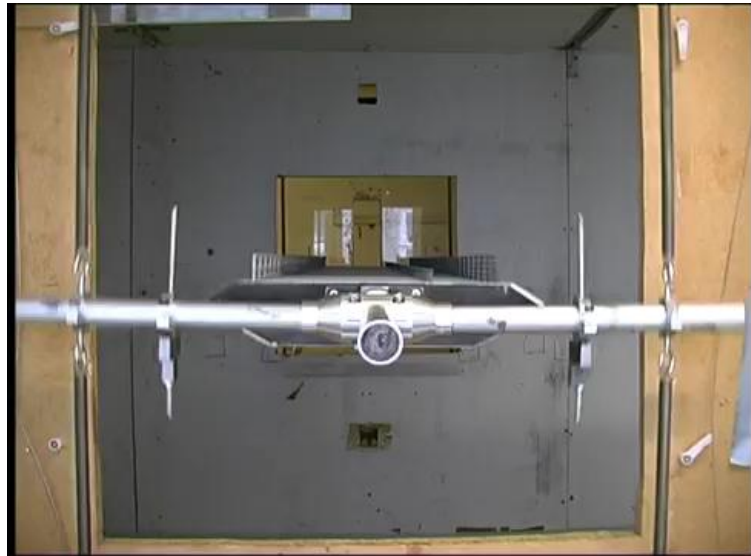
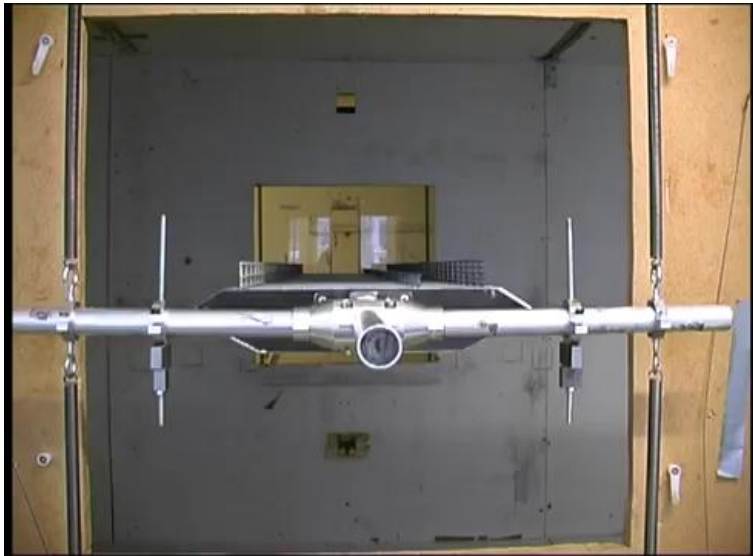


Holmes (2015)

Dynamic response - General

<https://www.newsflare.com/video/240089/weather-nature/video-shows-building-shaking-in-hong-kong-during-typhoon-mangkhut>

Dynamic response - General



Bridge deck tests @
Tongji University, China
Courtesy: Ting Yang

Dynamic response – small build./facades



<https://szavala1.files.wordpress.com/2011/01/ribbon-house-with-unusual-roof-1-554x369.jpg>



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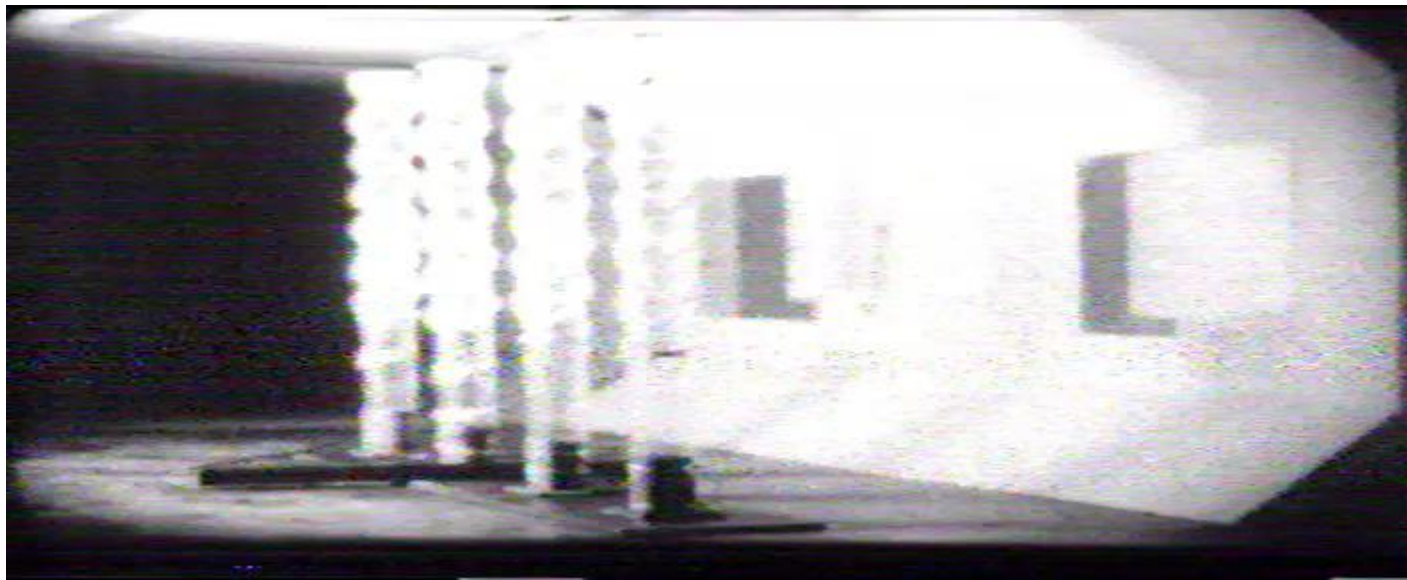


Dynamic response – Chimneys



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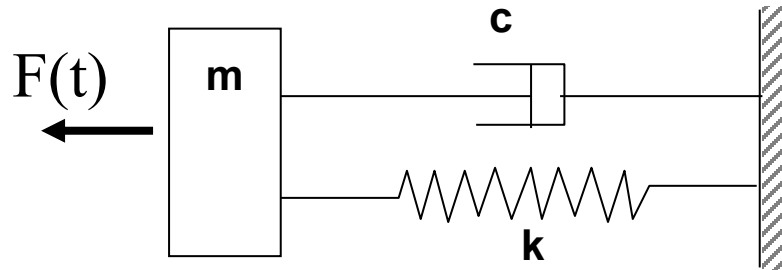


Dynamic response – non-structural members



Theory of vibration

Single degree of freedom system



$F(t)$: Excitation force
 m : mass of a system [kg]
 c : damping of a system [kg/s]
 k : stiffness of a system [N/m]
 $\ddot{x}, \dot{x}, x$: acceleration, velocity, displacement

$$F(t) = m\ddot{x} + c\dot{x} + kx$$

Theory of vibration

Undamped system

In the special case where damping $c = 0$, the eqn of motion becomes $0 = m\ddot{x} + kx$, when we set excitation to zero and solution can be shown to be:

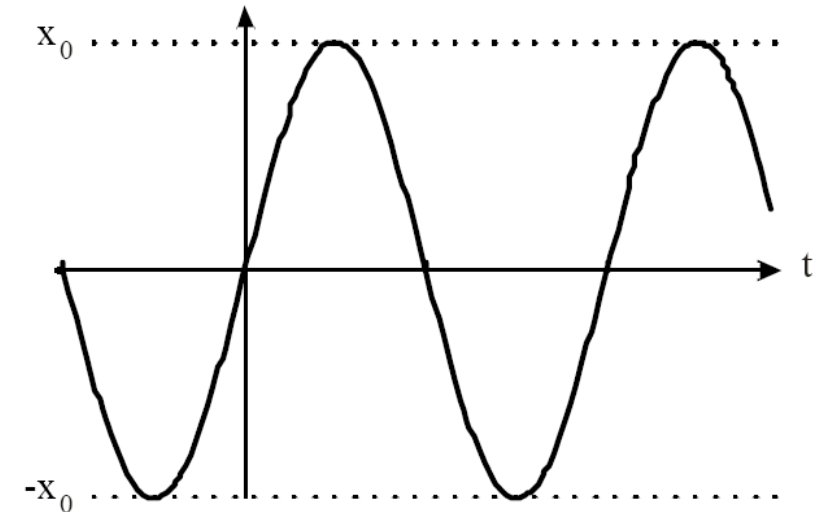
$$x(t) = A \cos(2\pi n_0 t) + B \sin(2\pi n_0 t)$$

Which has the derivative

$$v(t) = -A(2\pi n_0) \sin(2\pi n_0 t) + B(2\pi n_0) \cos(2\pi n_0 t)$$


$$n_0 = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$$

Undamped natural frequency



Giving the system an initial displacement and velocity then allows us to determine the constants A and B.

Theory of vibration

Damped vibration

$$F(t) = m\ddot{x} + c\dot{x} + kx$$

Solve by setting forcing equal to zero

Soln. has the form:

$$x(t) = Ae^{pt}$$

$$mp^2 + cp + k = 0$$

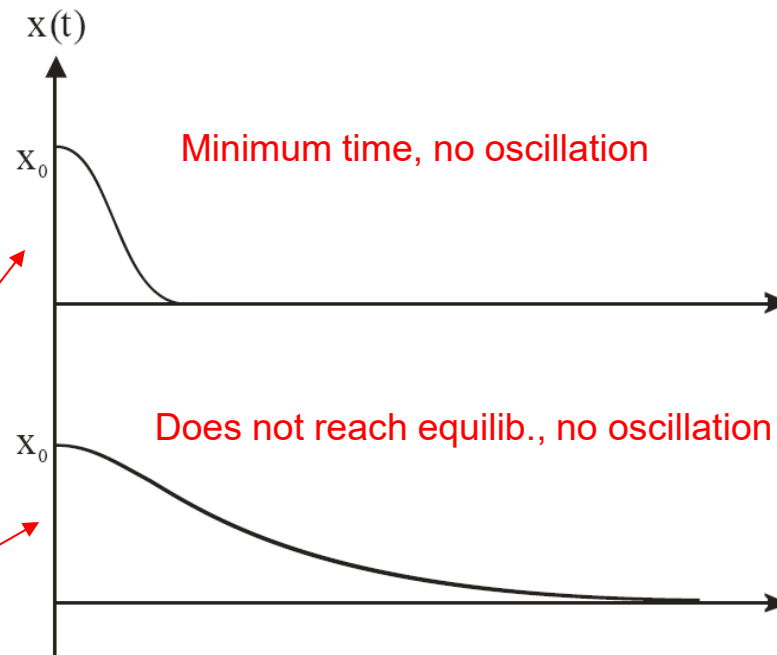
Where p is given by the roots of:

$$p = \frac{-c \pm \sqrt{c^2 - 4mk}}{2m}$$

~~$c^2 = 4mk \rightarrow$ critically damped~~

~~$c^2 > 4mk \rightarrow$ overdamped~~

$c^2 < 4mk \rightarrow$ underdamped



Not common in civil structures

Theory of vibration

- Underdamped ($c^2 < 4mk$)

Again undergoes oscillation, so cos and sine functions again work as a solution.

$$x(t) = e^{-2\pi n_0 \zeta t} (A_1 \cos(2\pi n_1 t) + A_2 \sin(2\pi n_1 t))$$

Where: $\zeta = \frac{c}{c_c}$

Damping ratio (points to ζ)

Damping (points to c)

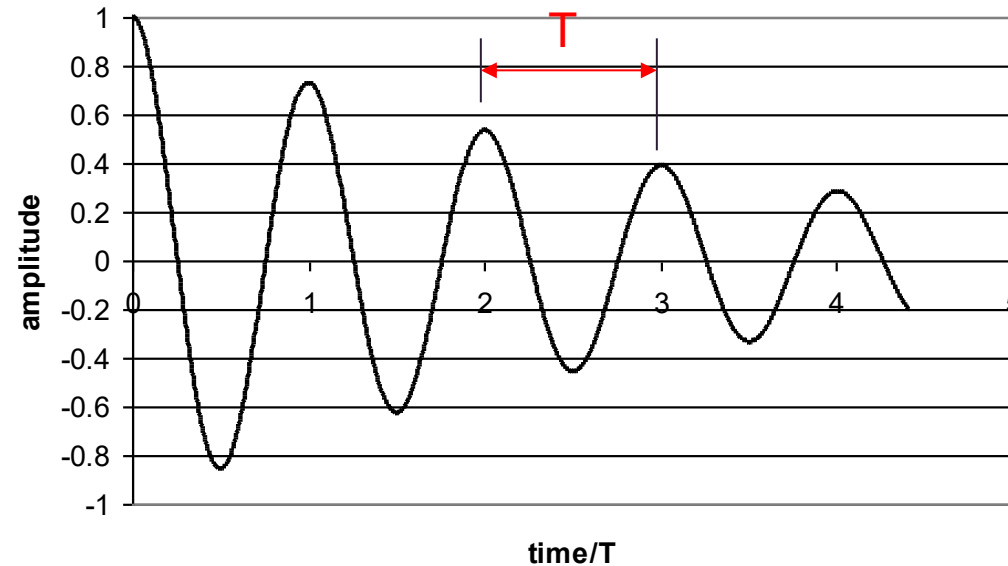
Critical damping (points to c_c)

1st mode nat. freq. (points to n_1)

$$n_1 = n_0 \sqrt{1 - \zeta^2}$$

$$T = \frac{1}{n_1} = \frac{1}{n_0 \sqrt{1 - \zeta^2}} \approx \frac{1}{n_0}$$

Damped period of vibration (points to T)



For most civil structure $\zeta = 0.005-0.05$

Response to sinusoidal loading

$$F_0 \sin(2\pi n t) = m\ddot{x} + c\dot{x} + kx$$

Steady state solution:

$$x(t) = (F_0/k) |H(n)| \sin(2\pi n t + \phi)$$

Static displacement

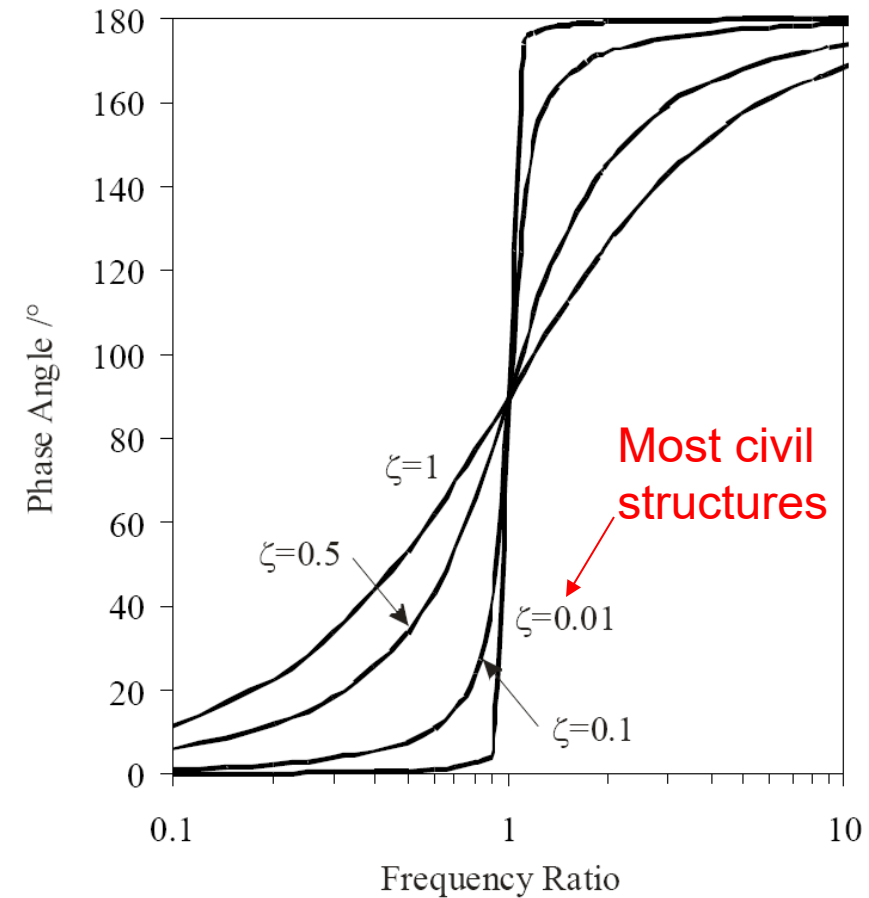
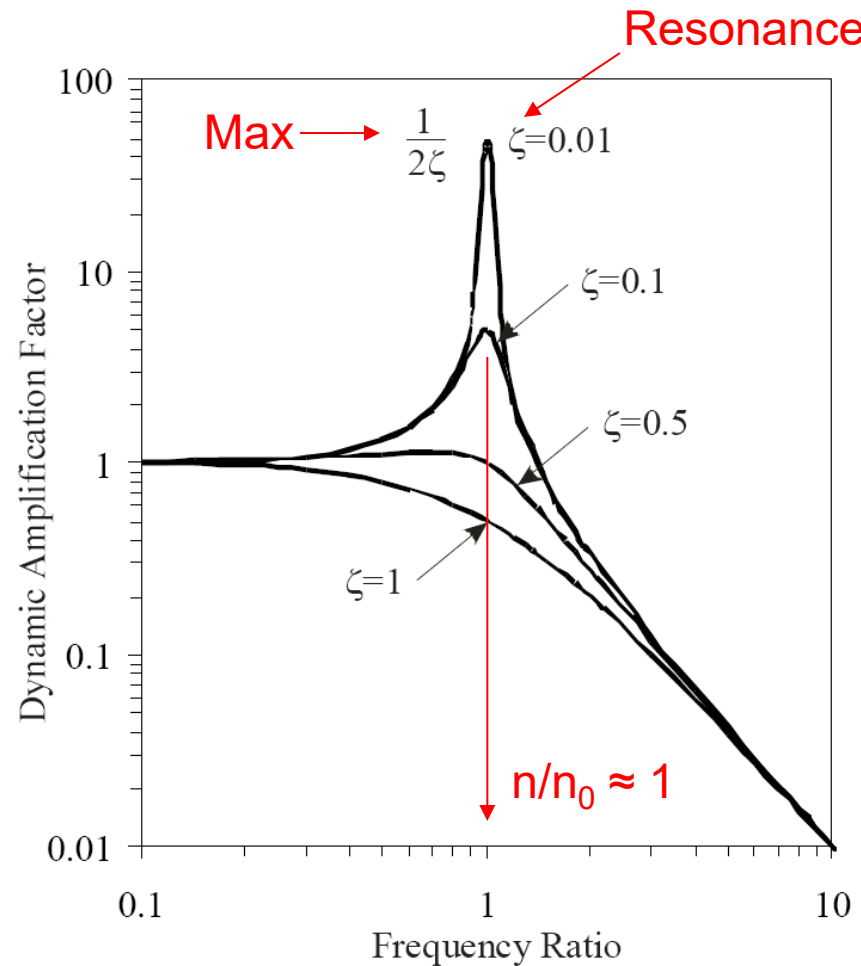
Dynamic amplification
(mechanical admittance)

Phase angle

$$|H(n)| = \frac{1}{\sqrt{\left[1 - \left(\frac{n}{n_0}\right)^2\right]^2 + 4\zeta^2 \left(\frac{n}{n_0}\right)^2}}$$

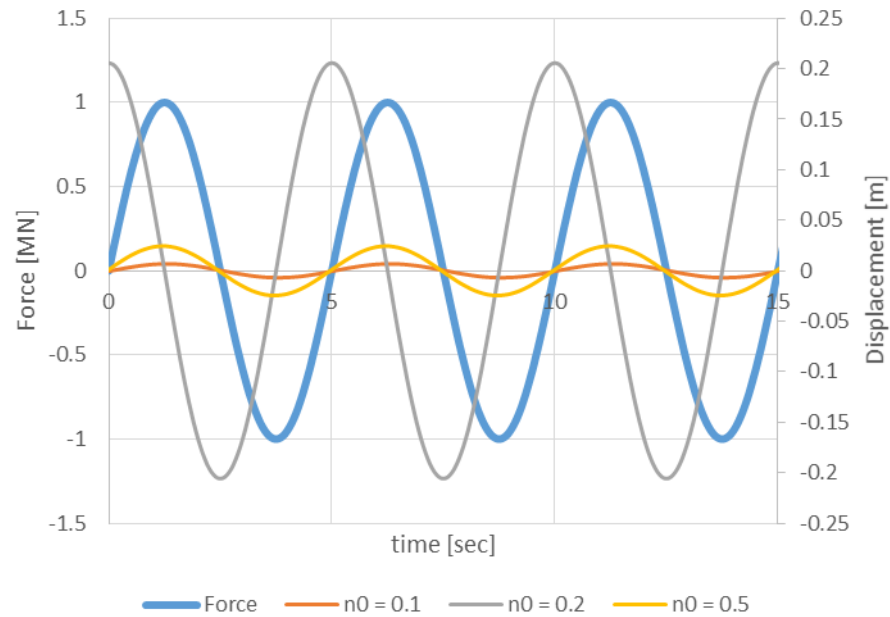
$$\phi = \tan^{-1} \left(\frac{2\zeta \frac{n}{n_0}}{1 - \left(\frac{n}{n_0}\right)^2} \right)$$

Response to sinusoidal loading

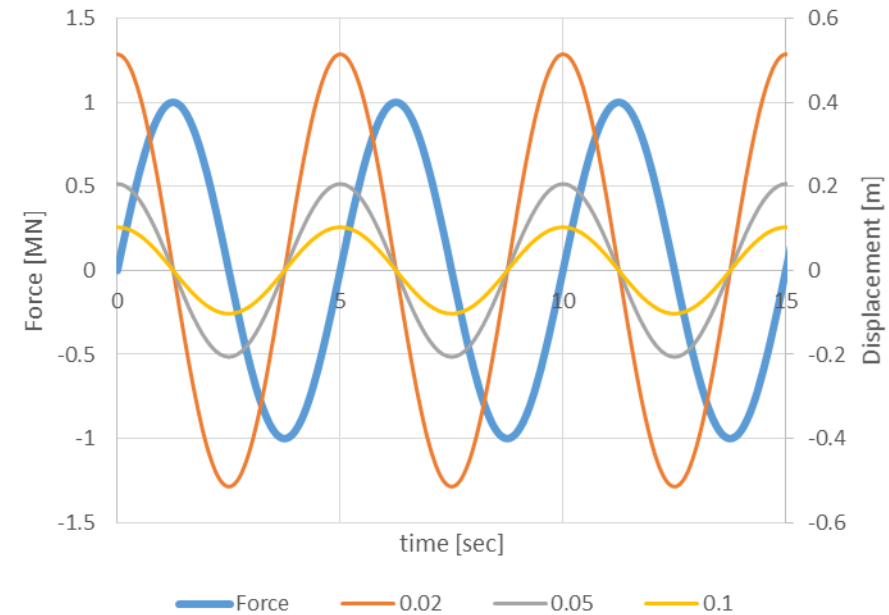


Response to sinusoidal loading

$n = 0.2$ (i.e. period = 5 sec)



Varying n_0

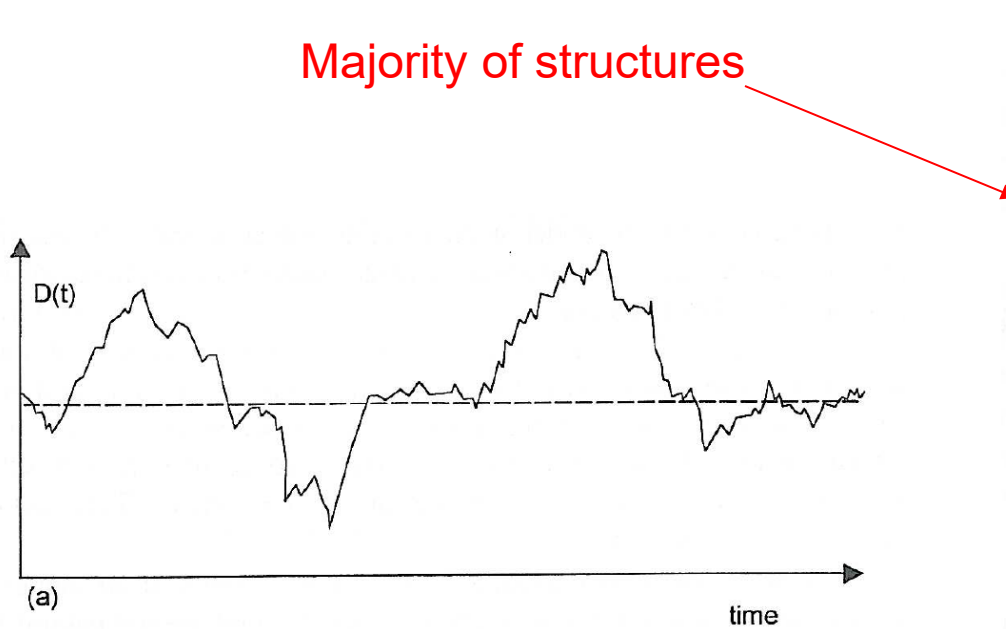


Varying damping

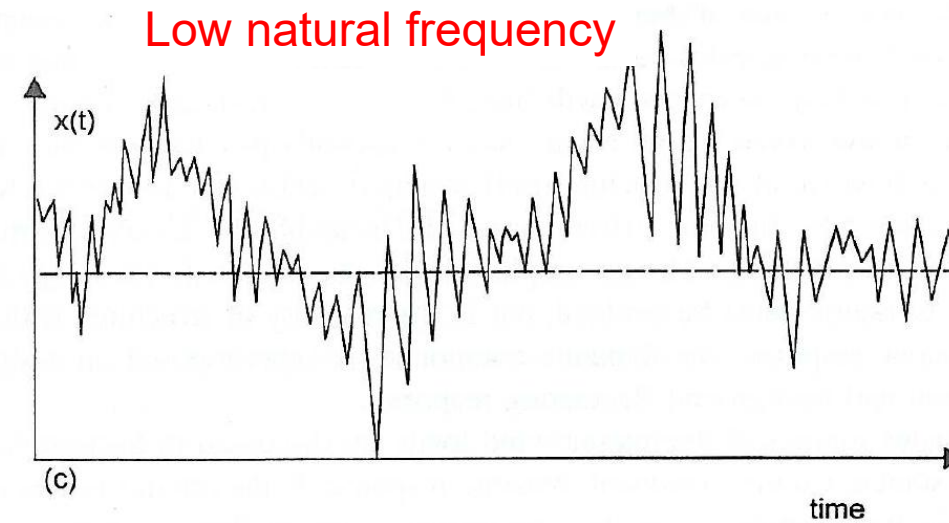
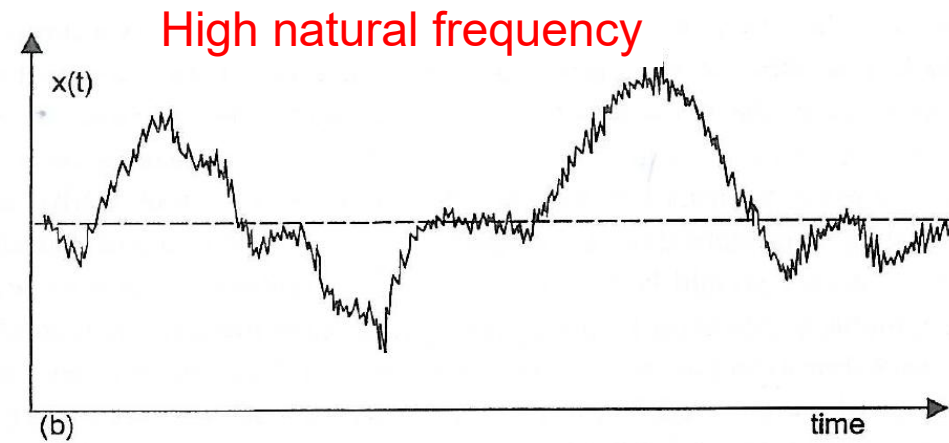
Response to wind

- Problem: Wind does not have one single forcing frequency, and it is not sinusoidal (but you can think of it as a series of them)!!
- Fluctuations in the wind are considered as a random process therefore the loading is also dealt with in this manner (frequency domain – spectral approach).
- Probabilistic, not deterministic – so can't recreate time history.
- Wind can potentially (rule of thumb) excite structures or components with $n_0 < 1$ Hz.

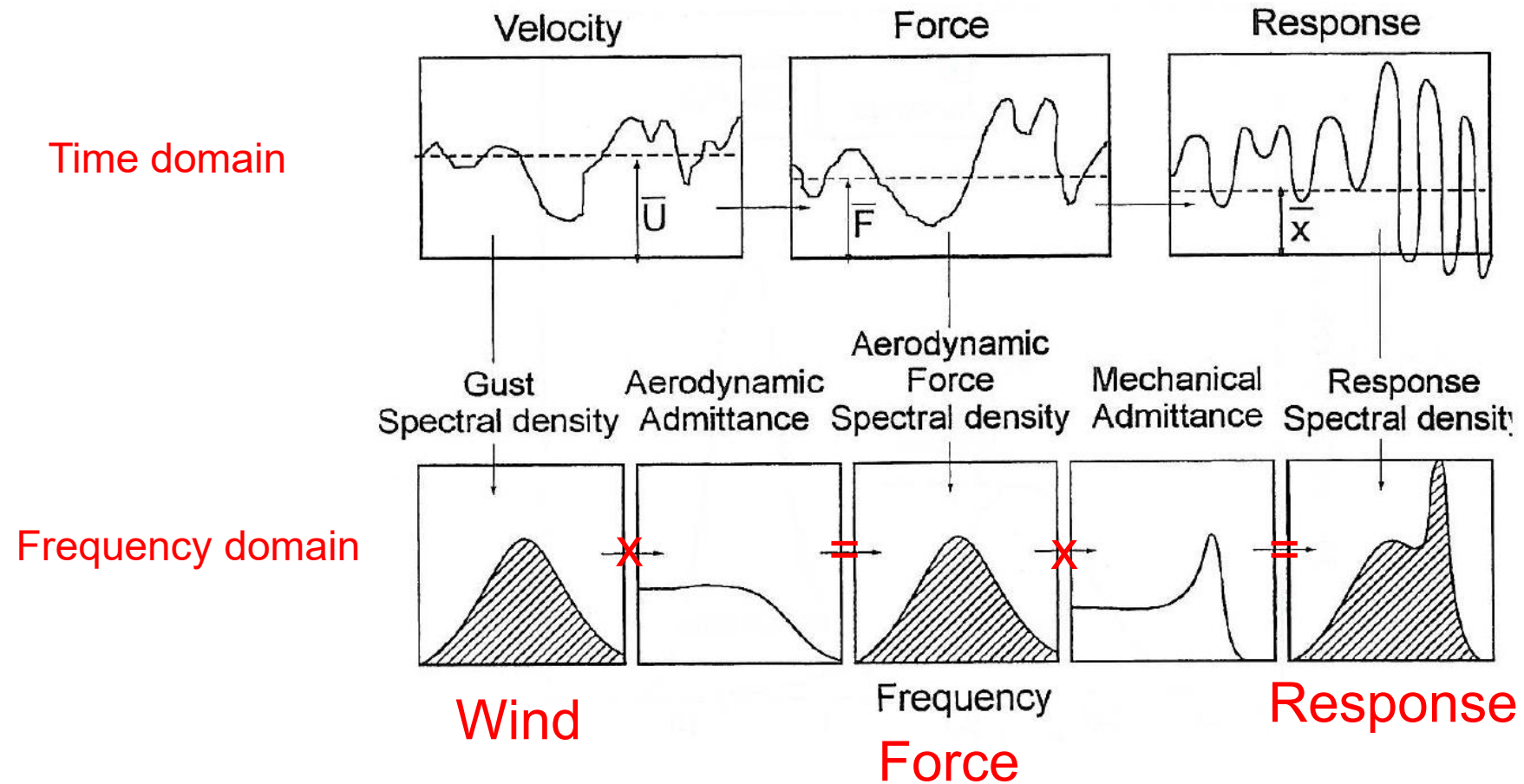
Response to random excitation (Wind)



Wind time history



Response to random excitation (Wind)



Random vibration

Consider an applied (random) wind force, $F(t)$, with the spectra $S_F(n)$

$$F(t) = m\ddot{x} + c\dot{x} + kx$$

$$S_x(n) = \left(\frac{1}{k}\right)^2 |H(n)|^2 S_F(n)$$

Response spectra

Dynamic amplification
squared

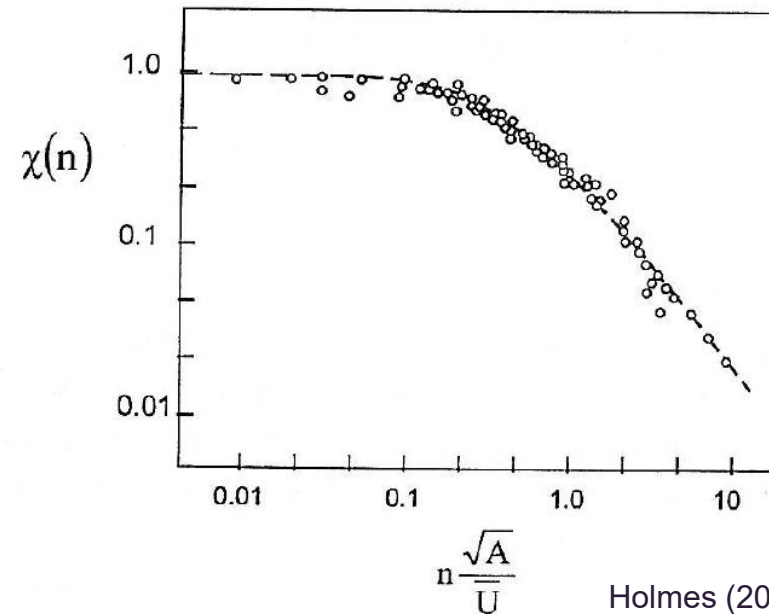
Wind force spectra
(not wind spectra)

Random vibration

For larger structures the load is not fully correlated

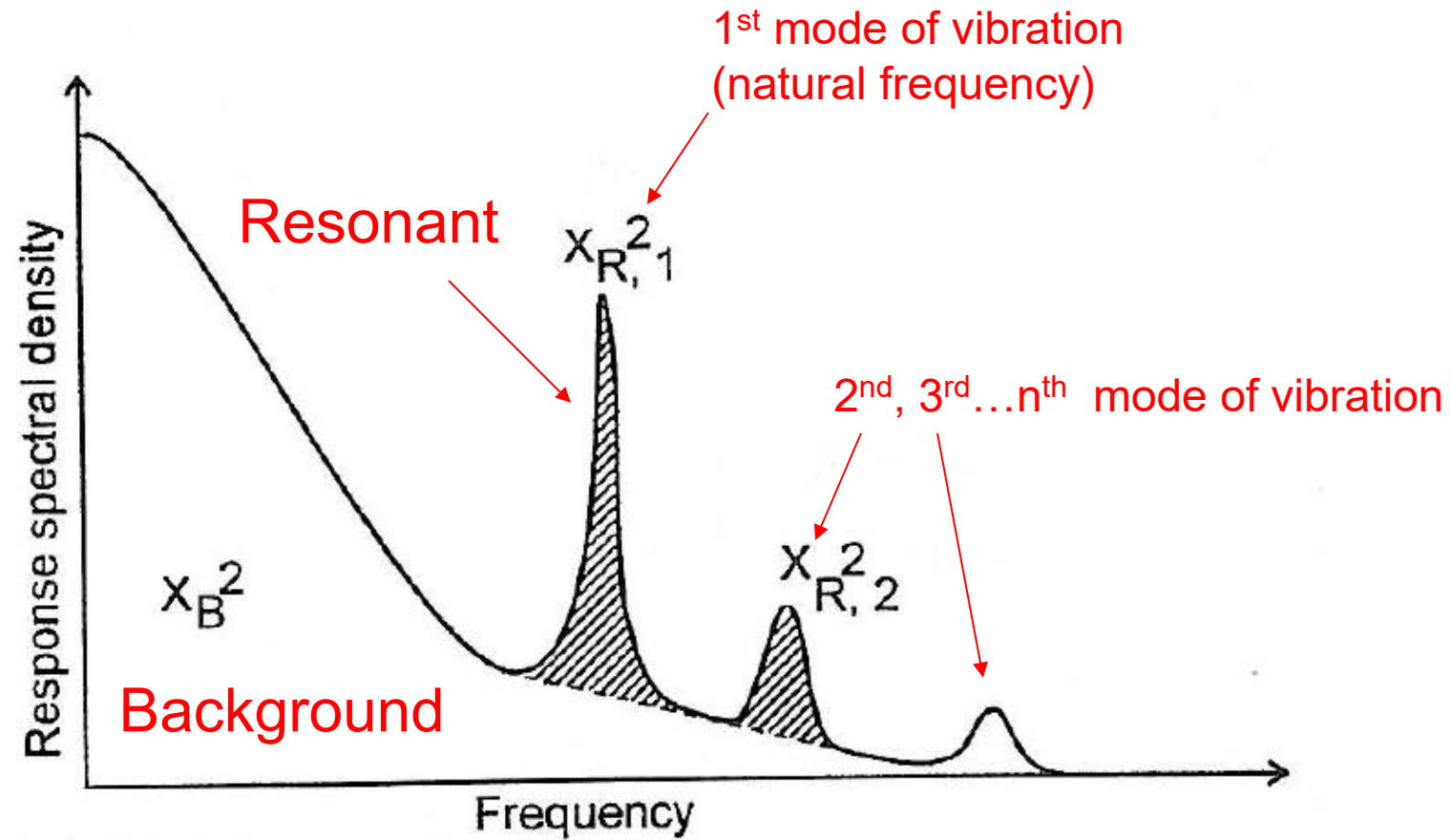
Aerodynamic admittance

$$S_x(n) = \left(\frac{1}{k}\right)^2 |H(n)|^2 \chi^2(n) S_F(n)$$



Holmes (2001)

Response spectra



Total response = Mean + Background + resonant

Random vibration

Area under spectra is variance

$$\sigma_x^2 = \int_0^{\infty} S_x(n) \, dn = (1/k)^2 \cdot \int_0^{\infty} |H(n)|^2 \cdot \chi^2(n) S_F(n) \, dn$$

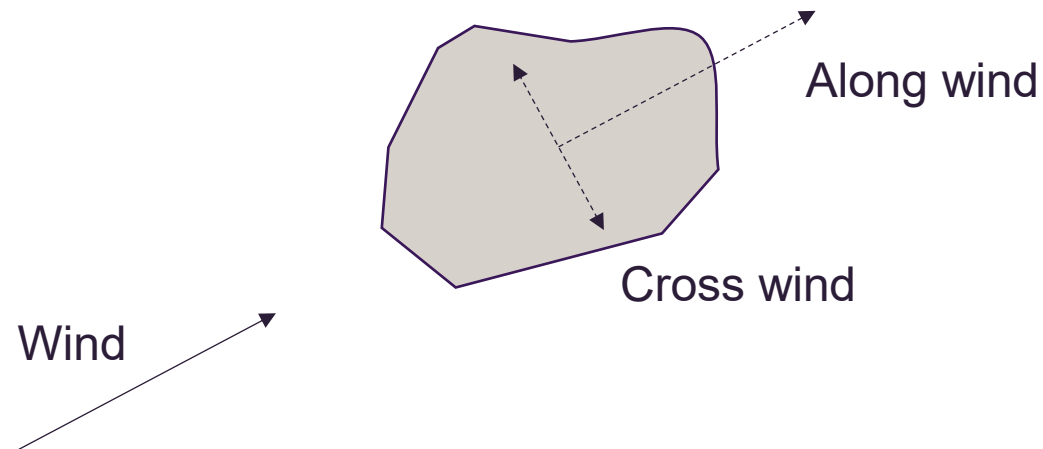
Can calculate peak response through a peak factor approach. This is useful for codification!

$$\hat{x} = \bar{x} + g\sigma_x$$

Response to wind

Given wind turbulence is three dimensional, dynamic response is also three dimensional.

Along wind response, cross wind response (possibly x 2), torsional response



Along wind response

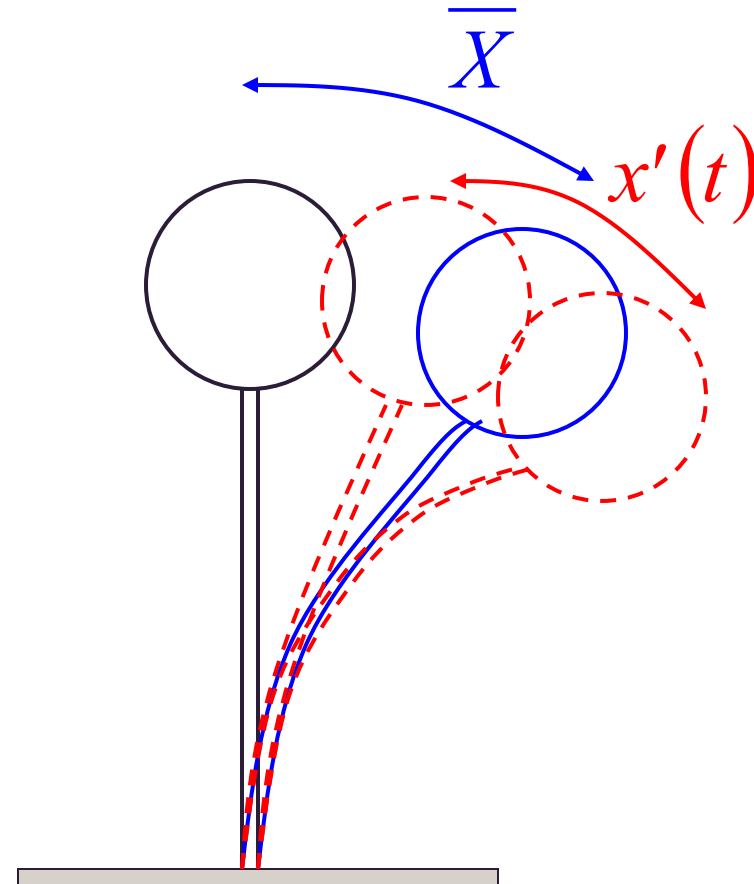
- Along wind dynamic response is directly related to oncoming turbulence - Buffeting.
- Effectiveness of a gust in loading a structure is dependent on its size compared to the size of the structure (particularly the windward face)

$$X(t) = \bar{X} + x'(t)$$

Wind speed
Wind pressure
Structural response

Mean

Fluctuating



Along wind response

Can directly apply random vibration theory to along wind loading of structures using previous equation.

$$S_x(n) = \left(\frac{1}{k} \right)^2 S_F(n) |H(n)|^2 \chi^2(n)$$

But we need to:

Relate wind spectra to wind force spectra

Determine mechanical admittance for each individual structure (generally requires wind tunnel testing)

Along wind response

Wind spectra → correlated gust force spectra

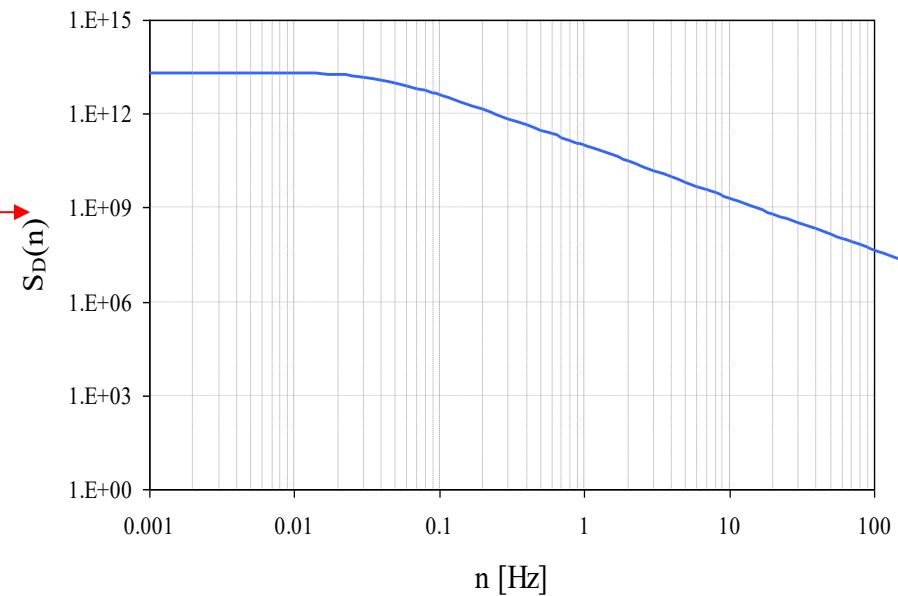
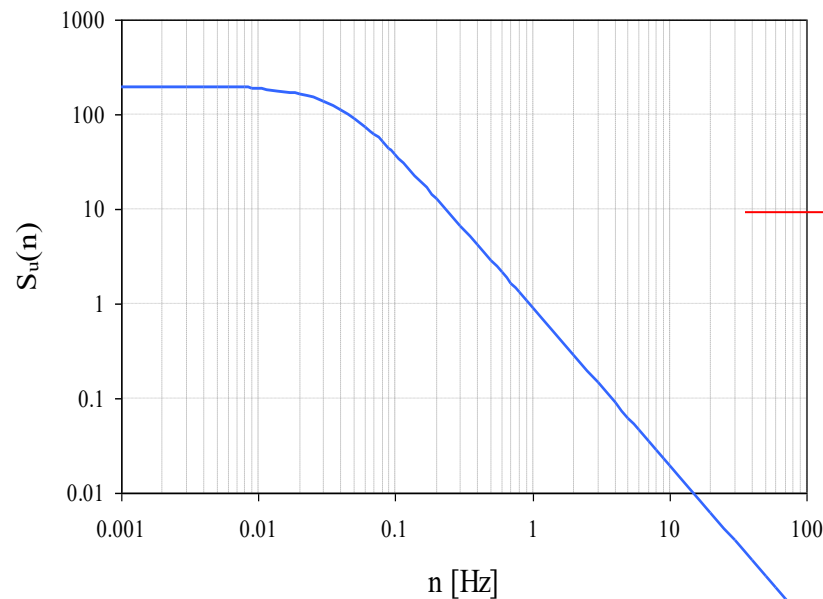
Theoretical, fully correlated wind force spectra →

$$S_D(n) = \frac{4\overline{F_D}^2}{\overline{U}^2} S_u(n)$$

Mean wind speed

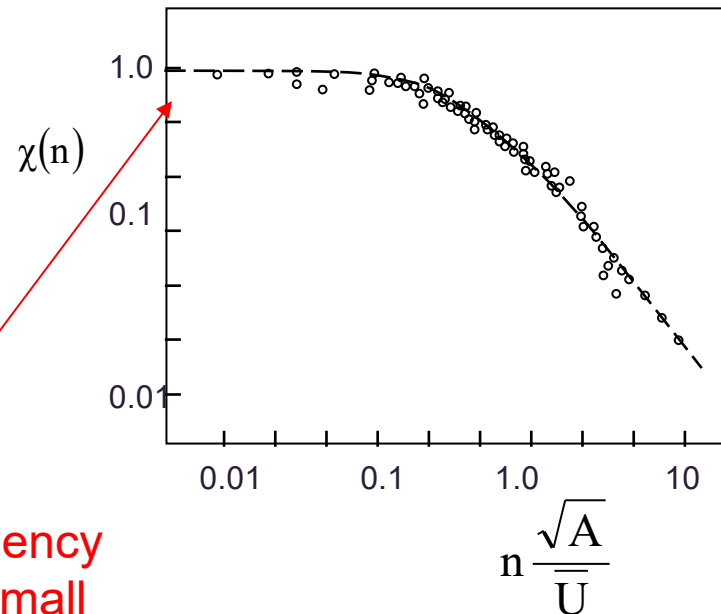
Mean along wind drag force

Wind spectra of fluctuating part



Along wind response

Aerodynamic admittance



Low frequency
gusts or small
structure – well
correlated

high frequency
gusts or large structure
– poorly correlated

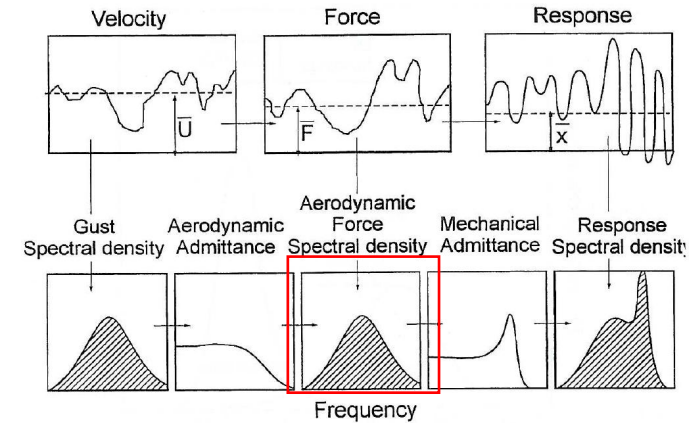
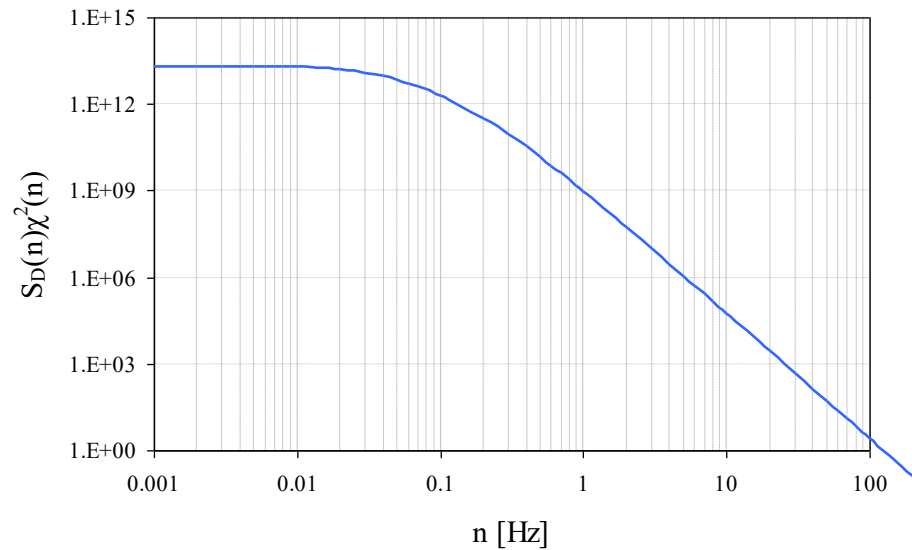
An appropriate form of χ
for plates and prisms is:

$$\chi(n) = \frac{1}{1 + \left[\frac{2n\sqrt{A}}{\bar{U}} \right]^{4/3}}$$

Along wind response

Non-correlated gust forcing spectra

$$S_D(n) = \frac{4\overline{F_D}^2}{U^2} S_u(n) \cdot \chi^2(n)$$



Along wind response

Subbing into the equation of motion

$$S_x(n) = \left(\frac{1}{k}\right)^2 |H(n)|^2 S_D(n)$$

Drag force spectra

$$S_x(n) = \left(\frac{1}{k}\right)^2 |H(n)|^2 \frac{4\overline{F_D}^2}{\overline{U}^2} S_u(n) \cdot \chi^2(n)$$

Relationship between force and displacement $\longrightarrow F_D = k \overline{X}$ Static displacement

$$S_x(n) = |H(n)|^2 \frac{4\overline{X}^2}{\overline{U}^2} S_u(n) \cdot \chi^2(n)$$

Along wind response

Area under spectrum = variance

$$\sigma_x^2 = \int_0^\infty S_x(n) dn = \int_0^\infty |H(n)|^2 \frac{4\bar{X}^2}{\bar{U}^2} S_u(n) \cdot \chi^2(n) \cdot dn$$

$$\sigma_x^2 = \frac{4\bar{X}^2 \sigma_u^2}{\bar{U}^2} \int_0^\infty |H(n)|^2 \cdot \chi^2(n) \cdot \frac{S_u(n)}{\sigma_u^2} \cdot dn \cong \frac{4\bar{X}^2 \sigma_u^2}{\bar{U}^2} [B + R]$$

Some assumptions need to be made

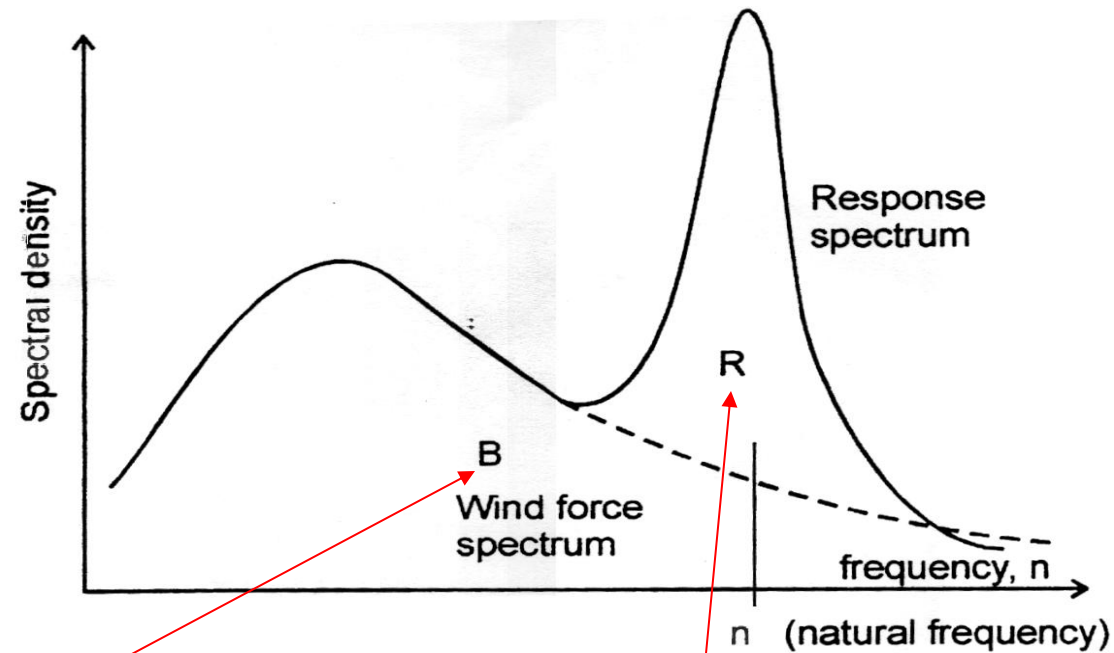
$$B = \int_0^\infty \chi^2(n) \cdot \frac{S_u(n)}{\sigma_u^2} \cdot dn$$

Background response factor

$$R = \chi^2(n_0) \cdot \frac{S_u(n_0)}{\sigma_u^2} \int_0^\infty |H(n)|^2 \cdot dn$$

Resonant response factor

Along wind response



$$B = \int_0^{\infty} \chi^2(n) \cdot \frac{S_u(n)}{\sigma_u^2} \cdot dn$$

$$R = \chi^2(n_0) \cdot \frac{S_u(n_0)}{\sigma_u^2} \int_0^{\infty} |H(n)|^2 \cdot dn$$

Along wind response

For codification a further assumption is made:

$$\int_0^{\infty} |H(n)|^2 \cdot dn = \frac{\pi n_0}{4\zeta}$$

Therefore:

$$R = \chi^2(n_0) \cdot \frac{S_u(n_0) \pi n_0}{4\sigma_u^2} \cdot \frac{1}{\zeta} \longrightarrow \sigma_x = 2\bar{X}I_u \sqrt{B + \frac{SE}{\zeta}}$$

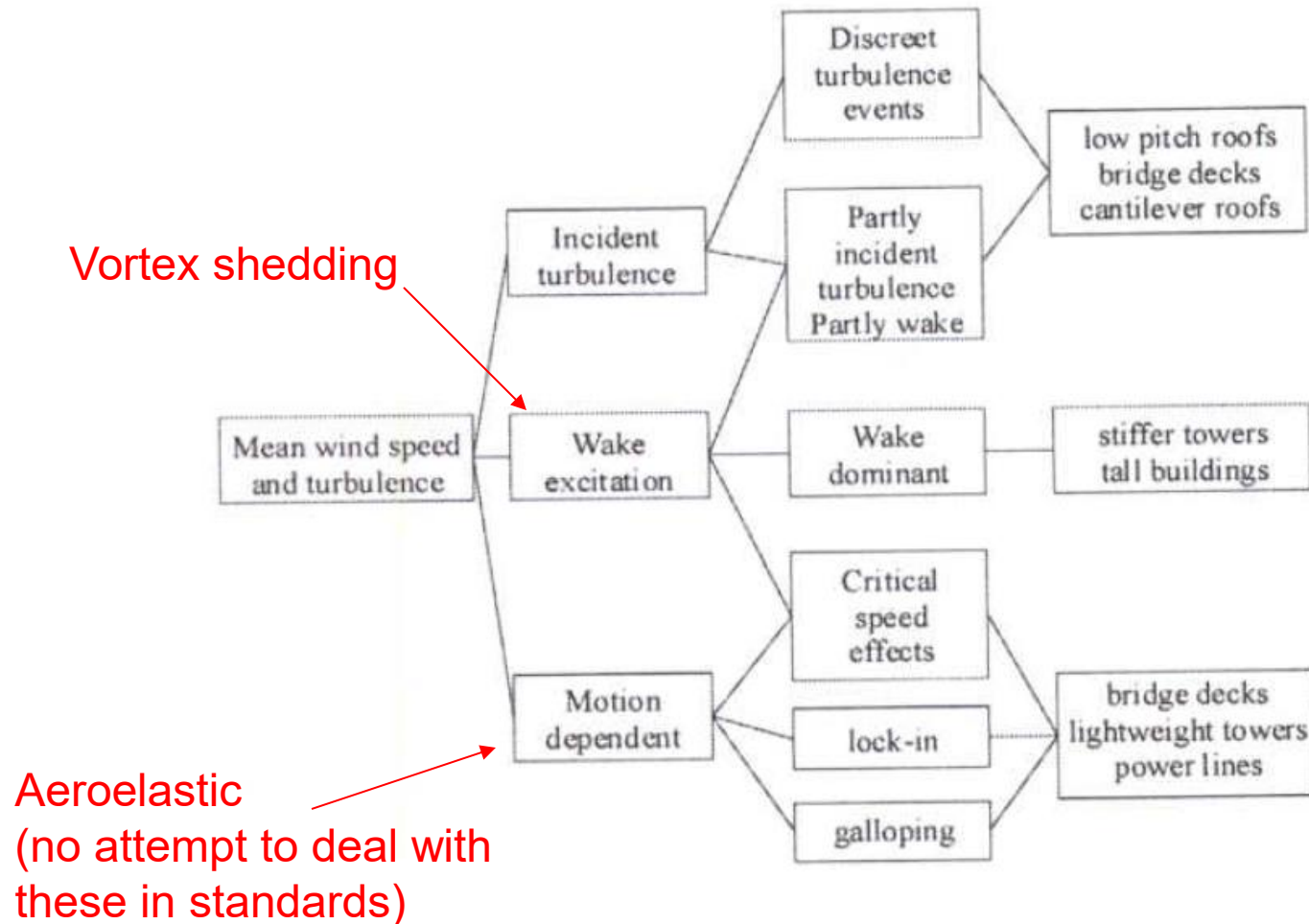
S
Size reduction
E
Energy in the wind

$$\hat{X} = \bar{X} + g\sigma_x$$

$$\hat{X} = \bar{X} + 2g\bar{X}I_u \sqrt{B + \frac{SE}{\zeta}} = \bar{X} \left[1 + 2gI_u \sqrt{B + \frac{SE}{\zeta}} \right] = G\bar{X}$$

Gust response factor

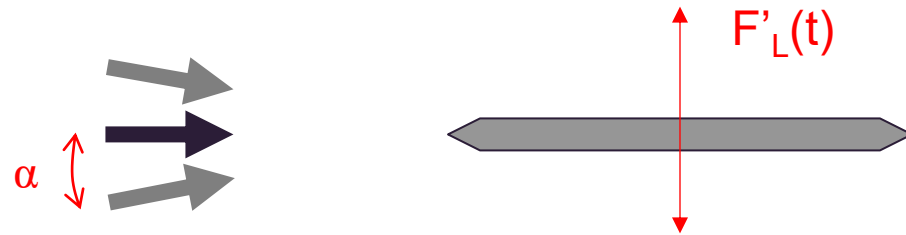
Cross wind response



Incident turbulence

Only significant for structures with large afterbody lengths (e.g. cantilever roofs, bridge decks).

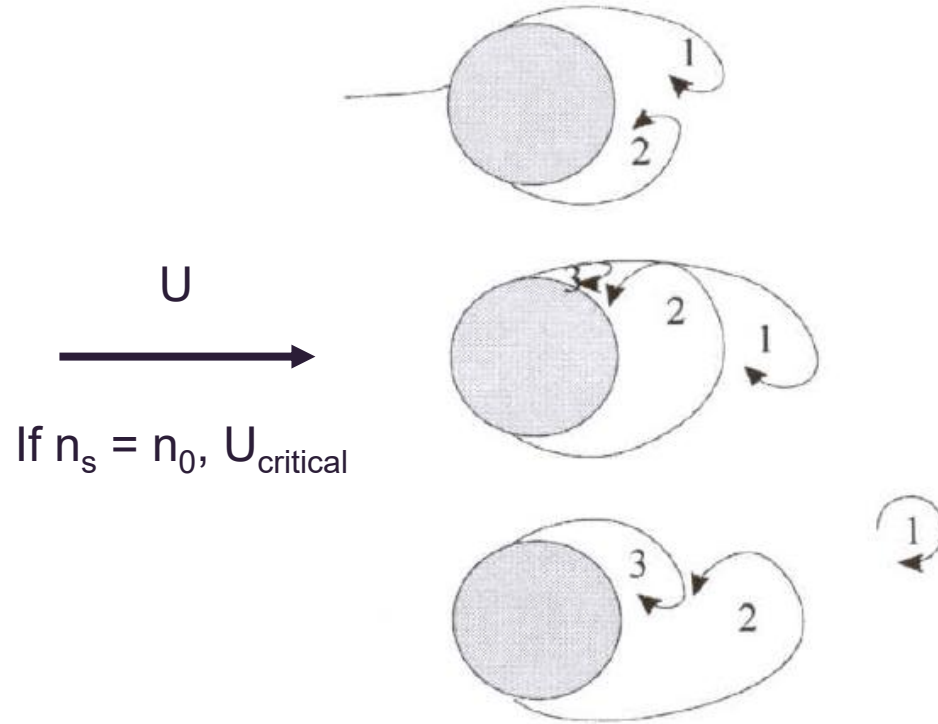
$$F'_L(t) = C_L \cdot \rho \cdot \bar{U} \cdot b \cdot u'(t) + \left(C_D + \frac{\partial C_L}{\partial \alpha} \right) \rho \cdot \bar{U} \cdot b \cdot v'(t)$$



Mean F_D and F_L are
small but $dC_L/d\alpha$ is large

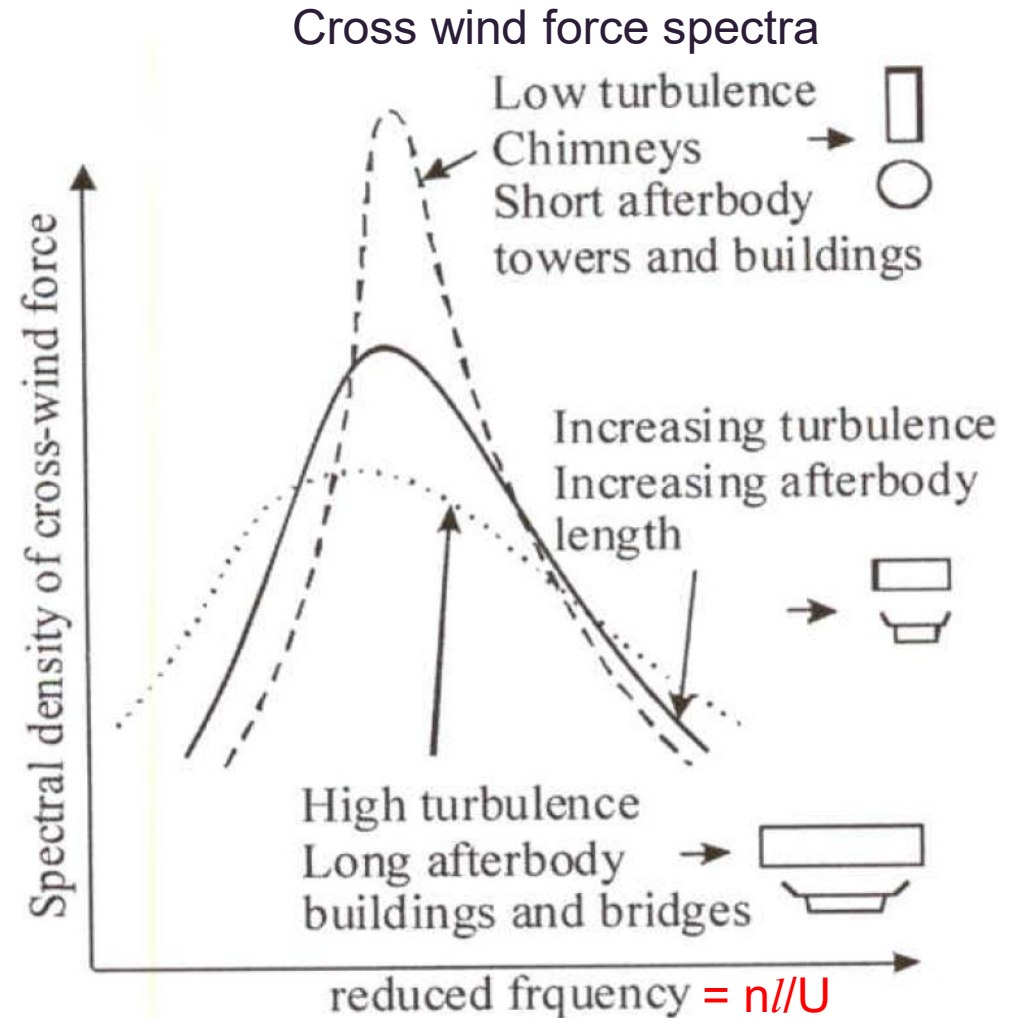
Wake excitation

Vortex shedding is most common.



Response spectra

- Increased turbulence decreases peak response and broadens spectra – reduced vortex shedding.
- Decrease in h/b ratio broadens spectra and reduces St .
- The longer the afterbody the broader the spectra.
- For tall buildings peak cross wind response is generally $>$ along wind – but not across all wind speeds.



Peak cross wind response

A Gaussian distribution is assumed:

Cross wind
lift force
spectra
(empirical)

Cross wind force coefficient

$$S_L(n) = \frac{(0.5 \cdot C_L \cdot \rho \cdot \bar{U}^2 \cdot b)^2}{B^* \cdot n_s \sqrt{\pi}} \cdot \exp\left(-\frac{1 - \frac{n}{n_s}}{B^*}\right)$$

Effective spectral bandwidth

Shedding frequency

**Replaces wind
spectra x
aerodynamic
admittance in
along wind**

Assuming the wake excitation is random:

$$\sigma_y^2 = \int_0^\infty S_y(n) dn = \frac{1}{(2\pi n_0)^4 m^2} \int_0^\infty S_L(n) \cdot |H(n)|^2 dn$$

$1/k^2$

Modal mass (if mass is distributed)

Peak cross wind response

As with the along wind, this can be broken into Background and Resonant components:

$$\sigma_y^2 \approx \frac{1}{(2\pi n_0)^4 m^2} \left[\int_0^\infty S_L(n) dn + S_L(n_0) \cdot \int_0^\infty |H(n)|^2 dn \right] \approx \frac{1}{(2\pi n_0)^4 m^2} \left[\int_0^\infty S_L(n) dn + \frac{\pi n_0 S_L(n_0)}{4\zeta_s} \right]$$

Background
response factor
Resonant
response factor

 $\int_0^\infty |H(n)|^2 .dn = \frac{\pi n_0}{4\zeta}$

Then making the assumption that the background response is small

$$\sigma_y \approx \frac{1}{(2\pi n_0)^2 m} \sqrt{\frac{\pi n_0 S_L(n_0)}{4\zeta_s}}$$

Wake excitation summary

- Derived equations are applicable for velocities below the critical velocity (ie velocity that induces wake excitation).
- Above critical velocity aeroelastic mechanisms occur, and predicted results underestimate actual response.

Dynamic response factor – C_{dyn} [6]

AS/NZS1170.2 Section 6

Limited scope

Tall buildings or freestanding towers

- $n > 1.0\text{Hz}$, $C_{dyn} = 1.0$
- $0.2\text{Hz} < n < 1\text{Hz}$, OK
- $n < 0.2\text{Hz}$, out of scope
- $h < 200\text{ m}$
- 1st and 2nd sway mode $> 10\%$ apart
- $\text{height}/\min(\text{width}, \text{breadth}) < 6$

Cantilever roofs

- $0.5\text{Hz} < n < 1\text{Hz}$
- Appendix D

Along wind response, crosswind response and combination of both.



Dynamic response factor – [6]

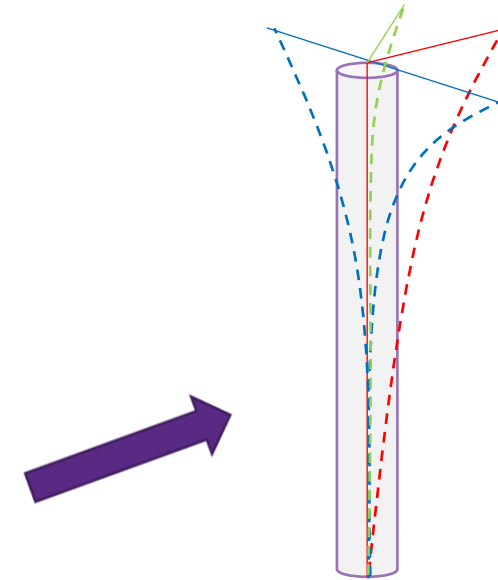
Along-wind response (C_{dyn})

Cross-wind response (Equivalent static force, base moments, tip deflection)

Combined response

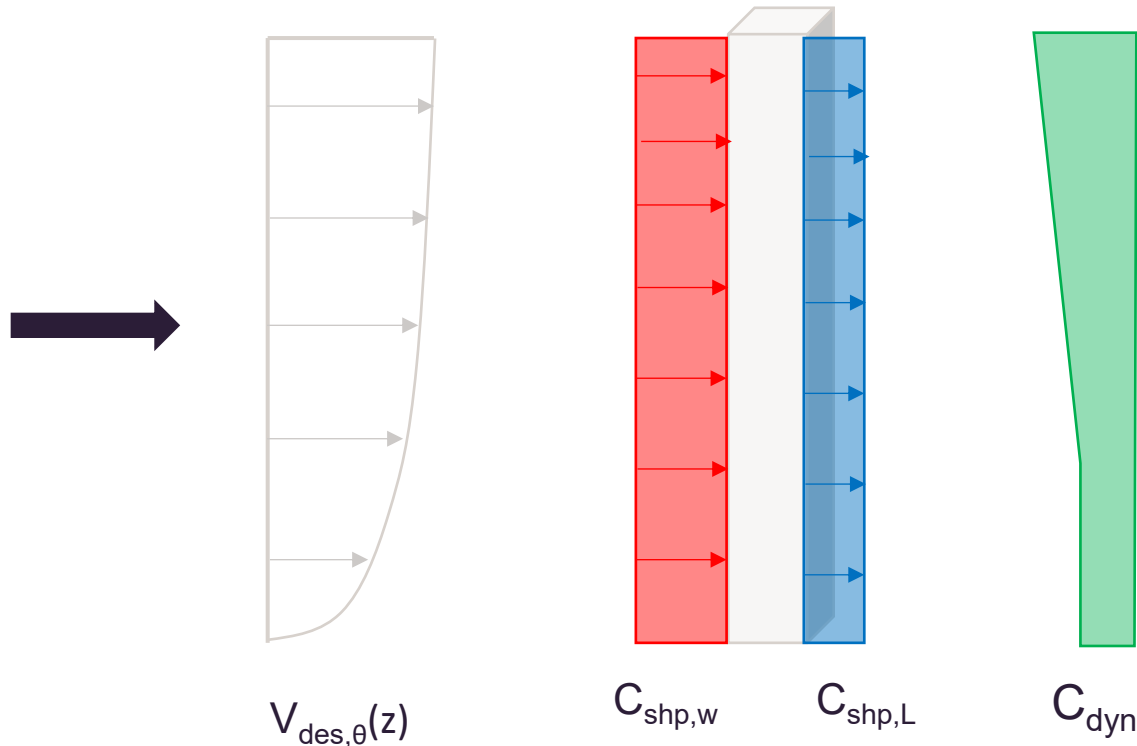
Dependent on:

- Shape
- Natural frequency
- Height, slenderness
- Damping
- Turbulence (wind speed)



Example

Calculate the dynamic response factors to apply to a 10 m tall reinforced concrete rectangular exhaust chimney. The chimney has an external dimension of 1m x 1m, a natural frequency of 0.5 Hz and is located in open terrain. The design wind speed is 50 m/s at 10 m elevation.



Example

Start with what we're given:

$$V_{\text{des}} = 50\text{m/s}, b = 1 \text{ m}, h = 10\text{m}, n_a = 0.5 \text{ Hz}$$

AS/NZS1170.2 [eqn 6.2(1)]

$$C_{\text{dyn}} = \frac{1 + 2I_h \sqrt{g_v^2 B_s + \frac{H_s g_R^2 S E_t}{\zeta}}}{(1 + 2g_v I_h)}$$

We can then find for the wind:

$$I_h = 0.183 \text{ [Table 6.1] - turbulence intensity}$$

$$g_v = 3.4 \quad [6.4.1] - \text{peak factor for wind}$$

$$L_h = 85 \text{ m} \quad [\text{eqn 6.2(3)}] - \text{length scale for wind}$$

$$E = 0.072 \text{ [eqn 6.2(6)] - Spectra value at } V_{\text{des}}$$

Example

We can then find for the structure:

- $g_r = 3.55$ [eqn 6.2(4)] – peak factor for structural response
- $\zeta = 0.03$ [6.4.2 notes, reinforced concrete] - % critical damping
- $S = 0.6$ [eqn 6.2(5)] – Size reduction factor (aero admittance)

At **each elevation** we can then calculate

- $B_s(z=5m) = 0.97$ [eqn 6.2(6)] – Background response factor
- $B_s(z=10m) = 0.99$ [eqn 6.2(6)] – Background response factor
- $H_s (@5m) = 1.25$ [6.4.2] – height factor for resonance
- $H_s (@10m) = 2$ [6.4.2] – height factor for resonance

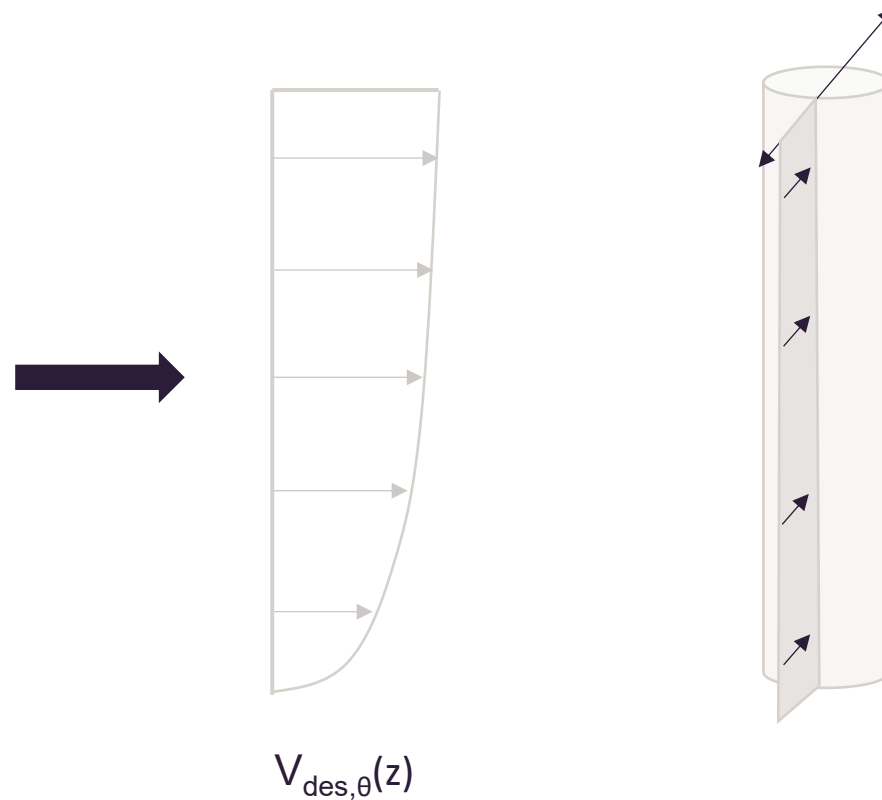
Then finally subbing back into the original eqn

- $C_{dyn}(z=5m) = 1.4$
- $C_{dyn}(z=10m) = 1.57$

$$C_{dyn} = \frac{1 + 2I_h \sqrt{g_v^2 B_s + \frac{H_s g_R^2 S E_t}{\zeta}}}{(1 + 2g_v I_h)}$$

Example 2

Assume now that the chimney is circular ($D = 1\text{m}$), calculate the equivalent static crosswind forces up the structure and the peak lateral (crosswind) deflection. All other properties stay as in Example 1.



Example 2

Unlike for along wind, C_{shp} is calculated with C_{dyn} in section 6

Force per
unit height

$$w_{eq}(z) = 0.5 \rho_{air} [V_{des, \theta}]^2 d C_{shp} C_{dyn}$$

[6.3(1)]



Depth

$$w_{eq}(z) = 0.5 \rho_{air} [V_{crit}]^2 b_t C_{shp} C_{dyn}$$

[6.3.(12)]



Critical velocity $\sim 5n_1 b_t$

Avg. breadth over top third

Also, for chimneys the critical velocity is (implicitly) used rather than the design wind speed. The procedure then becomes:

Max tip displacement

$$y_{max} = g_c \sigma_{y, max}$$

$$\left(\frac{\sigma_{y, max}}{b_t} \right)^2 = c_1 + \sqrt{c_1^2 + c_2}$$

$$w_{eq}(z) = m(z) (2\pi n_1)^2 y_{max} \phi_1(z)$$

Mass per unit height over top third

Example 2

Calculating first the maximum tip deflection

$b_t = 1 \text{ m}$ breadth of structure

$M(z) = 290 \text{ kg/m}$ (assuming $2,400 \text{ kg/m}^3$ for concrete and 40 mm walls)

....

[6.5.3.1]

....

$y_{\max} = 8.3 \text{ mm}$

Then to get the equivalent static force: $w_{\text{eq}}(z) = m(z) (2\pi n_1)^2 y_{\max} \phi_1(z)$

$w_{\text{eq}}(z) = 290 * (2*\pi*0.5)^2 * 0.0083 * (z/10)^2$

$w_{\text{eq}}(10\text{m}) = 23.7 \text{ Pa}$ tiny? ($v_{\text{crit}} \sim 5n_1b_t = 5*0.5\text{hz}*1\text{m} = 2.5 \text{ m/s}$)

Combined response

Combination of base moments

Table 6.3(A) — Load combinations for equivalent static loads for buildings < 70 m

Load cases	Peak load combinations
1 and 2	1.0(along-wind) ± 0.8(crosswind)
3 and 4	0.8(along-wind) ± 1.0(crosswind)

Table 6.3(B) — Load combinations for equivalent static loads for buildings > 70 m

Load cases	Peak load combinations
1 to 4	1.0(along-wind ± 0.15B offset) ± 0.8(crosswind)
5 to 8	0.8(along-wind ± 0.15B offset) ± 1.0(crosswind)
9 to 12	0.8(along-wind ± 0.2B offset) ± 0.8(crosswind)

Combination of load effects (e.g. forces in members)

$$\varepsilon_t = \varepsilon_{a,m} + \sqrt{(\varepsilon_{a,p} - \varepsilon_{a,m})^2 + \varepsilon_{c,p}^2} \quad 6.4(1)$$

where

$\varepsilon_{a,m}$ = action effect derived from the mean along-wind response, given as follows, where the values of g_v , I_h and C_{dyn} are defined in Clause 6.2.1:

$$\varepsilon_{a,p} / [C_{dyn} (1 + 2g_v I_h)]$$

$\varepsilon_{a,p}$ = action effect derived from the peak along-wind response

$\varepsilon_{c,p}$ = action effect derived from the peak crosswind response

NOTE 1 The factor $[C_{dyn} (1 + 2g_v I_h)]$ is a gust factor (G).

NOTE 2 Maximum action effects derived from the crosswind response of chimneys, masts and poles of circular cross-section (Clause 6.3.3), which occur at the critical wind speed for vortex shedding, should not be combined with action effects for along-wind response calculated at a different wind speed.

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